When psychopathic traits are not traits: Aggregation bias and Big Five redundancy in the prediction of professional success*

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Abstract

We replicate and extend Eisenbarth et al. (2018), who examine whether psychopathic traits predict professional success on a sample of $N = 439$ working adults (OSF data). The original zero-order correlation between Fearless Dominance (FD) and Professional Satisfaction ($r = +0.13$) does not survive once three methodological problems are addressed jointly. First, the PPI-R total score conflates the positive effect of FD with the negative effect of Self-Centered Impulsivity (SCI): the aggregated correlation is $r = -0.03$, hiding an opposite-sign component structure. Second, FD is largely captured by Big Five Extraversion and Emotional Stability: a regression of FD on the Big Five plus demographics yields $R^2 = 0.616$ ($N = 409$), 87.5% of which is recovered by Extraversion and Emotional Stability alone; the linear ranking of predictors and the information-theoretic ranking from the Kraskov–Stögbauer–Grassberger (KSG) mutual information estimator agree on all five positions, and the corrected redundancy ratio $1 - I(FD; BF)/H_{diff}(FD)$ equals 0.55 — the Big Five accounts for 45% of FD's information content, the residual 55% being idiosyncratic variance or measurement noise. Third, three causal estimators (caliper propensity-score matching, inverse probability weighting, and augmented IPW) applied at four treatment thresholds (FD > P50, P67, P75, P80) all return ATTs not significant at the 5% level. The caliper PSM at the median threshold drops 117 of 198 treated observations (59%) for lack of admissible matches within $0.2 \cdot SD(\text{logit } PS)$ — itself a structural symptom of redundancy: high-FD individuals have few comparable controls because high FD is essentially high Extraversion combined with high Emotional Stability. The methodological contribution is to show that aggregation bias and construct redundancy jointly explain previously reported FD effects: studies reporting an FD effect on professional outcomes without controlling for the Big Five are likely measuring Extraversion and Emotional Stability under a different label.

1 Introduction

Eisenbarth et al. (2018) ask whether psychopathic traits, as measured by the Lilienfeld and Widows (2005) Psychopathic Personality Inventory–Revised (PPI-R), predict professional success in a sample of working adults. They report a positive zero-order correlation between Fearless Dominance (FD) and Professional Satisfaction ($r = 0.12$, $p < 0.01$ in the original) that attenuates — but survives — the inclusion of Big Five controls. Their study is part of a broader literature documenting positive associations between "boldness" or "successful psychopathy" and career outcomes, with implications for labor economics narratives in which dominance traits are economically rewarded.

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This paper revisits the same data and finds that the FD effect on professional success rests on three methodological foundations that do not jointly hold.

The *first* problem is aggregation bias. The PPI-R total score sums three orthogonal dimensions: FD (positive correlate of Professional Satisfaction in our replication, $r = +0.13$), SCI (negative correlate, $r = -0.14$), and Coldheartedness (CO; null correlate, $r = -0.05$). The aggregate correlation is $r = -0.03$, a spurious null produced by the cancellation of opposite-sign component effects. Reading the aggregated score as a meaningful construct mistakes a measurement artefact for the absence of a relationship.

The *second* problem is construct redundancy. FD is empirically close to a linear combination of Big Five Extraversion and Emotional Stability: the bivariate correlations are $r = +0.49$ and $r = +0.66$ respectively, and the regression FD $\sim$ Big Five + demographics delivers $R^2 = 0.616$. A reduced model with only Extraversion and Emotional Stability captures 87.5% of that explanatory power. A nonparametric information-theoretic check (Kraskov–Stögbauer–Grassberger mutual information, Kraskov et al., 2004) confirms this: the joint mutual information $I(\mathrm{FD}; \mathrm{BF}_{\mathrm{full}}) = 0.93$ bits accounts for approximately 45% of the differential entropy of FD, with Extraversion and Emotional Stability dominating the contribution and no detectable nonlinear residual. The linear ranking of Big Five predictors and the information-theoretic ranking match position-by-position (5/5).

The *third* problem is the absence of a robust causal effect. Under three estimators (caliper propensity-score matching, inverse probability weighting, and augmented IPW following Robins et al., 1995) and four candidate dichotomisations of FD (P50, P67, P75, P80), no Average Treatment Effect on the Treated is significant at the 5% level. The caliper PSM at the median threshold, with caliper width $0.2 \cdot \mathrm{SD}(\text{logit PS})$ in line with the recommendation of Austin (2011), drops 117 of 198 treated units for lack of admissible matches: the failure of common support is itself a manifestation of redundancy, since high-FD individuals have few control units that combine comparable Extraversion and Emotional Stability profiles.

We make three contributions. First, a methodological diagnosis: the three problems above are not independent. Construct redundancy *causes* the common-support failure of PSM; aggregation bias *motivates* the disaggregated analysis whose null result triggers the redundancy diagnostic; the null causal estimates close the loop. Second, an honest reporting of effect sizes: the disaggregated zero-order correlations of FD and SCI with Professional Satisfaction are non-trivial, but their predictive R^2 in cross-validation is essentially zero across both linear (OLS) and nonlinear (XGBoost) specifications. Third, a corrected redundancy ratio for the FD–Big-Five relationship that combines the continuous KSG mutual information with the Gaussian differential-entropy normalisation, avoiding the unit-mismatch that occurs when one mixes continuous mutual information with discrete entropy of a binned variable.

The paper is organised as follows. Section 2 describes the dataset. Section 3 presents the five methodological pipelines: replication, factor structure, predictive modelling, causal identification, and redundancy diagnostic. Section 4 reports the empirical results in the same order. Section 5 discusses why the three problems are connected and what the implications are for the psychopathy and labor literatures. Section 6 states limitations and threats to identification. Section 7 concludes.

2 Data

The data is the working sample from Eisenbarth et al. (2018), made publicly available via the Open Science Framework (`https://osf.io/tgujv`). It consists of self-report responses from $N = 439$ working adults after the original exclusion criteria (failure to complete all measures, prior participation, age below 18, non-English first language). The dataset includes:

- **Psychopathic traits**: the 40-item short form of the PPI-R (Lilienfeld and Widows, 2005), scored into three dimensions — Fearless Dominance (FD), Self-Centered Impulsivity (SCI), and Coldheartedness (CO) — and a total score (PPI Total).
- **Big Five personality**: the Ten-Item Personality Inventory (Gosling et al., 2003), scoring Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness.
- **Professional Satisfaction**: a composite of three items (Career Satisfaction, Promotion Satisfaction, Salary Satisfaction).
- **Material Success**: a composite indicator combining professional standing, annual salary band, and promotion frequency.
- **Demographics and controls**: age (mean 33.0, SD 9.2), gender (women 59.7%, men 40.3%), and months in current job.

For all analyses except those requiring complete cases on Months in Job, the sample is $N = 439$. The causal-inference and redundancy analyses condition on $N = 409$ after dropping rows with missing covariates (most commonly Months in Job).

A caveat applies to the Material Success composite. Cronbach's α is 0.48 in the original and 0.48 in our replication; this is below conventional reliability thresholds for psychometric use ($\alpha \geq 0.70$). We retain Material Success in the descriptive analysis (Section 4.1 and Table 2) for direct comparison with Eisenbarth et al. (2018) but treat Professional Satisfaction ($\alpha = 0.78$) as the primary outcome for the predictive, causal, and redundancy analyses. This reduces Section 4 to a single outcome at the cost of partial coverage of the original research question.

3 Methodology

3.1 Replication pipeline

We replicate the original analysis in three steps: (i) Cronbach's α for the Professional Satisfaction and Material Success composites and for PPI-R Coldheartedness (Cronbach, 1951); (ii) zero-order Pearson correlations among PPI dimensions, Big Five traits, and the two outcome composites (the original Table 3); and (iii) OLS regressions of each outcome on the three PPI dimensions, first without controls and then with Big Five and demographic controls (the original Table 4 estimated structurally as SEM, here estimated by OLS as a transparent approximation).

3.2 Factor structure and aggregation diagnostic

We perform an Exploratory Factor Analysis on the 40 PPI-R items using oblimin rotation, after verifying sampling adequacy via Bartlett's test of sphericity ($\chi^2 = 4{,}821.99$, $p < 0.001$) and the Kaiser–Meyer–Olkin measure (KMO = 0.807, in the "meritorious" range). The objective is to confirm the multidimensional structure of the inventory empirically.

The aggregation diagnostic is purely descriptive: we compute the zero-order correlation of each PPI dimension and of the PPI Total score with each outcome composite, and visualise the cancellation in Figure 2.

3.3 Predictive modelling

We compare four feature sets — PPI Total only; FD+SCI+CO; FD+SCI+CO plus Big Five; full model including demographics — under two estimators: ordinary least squares (OLS) and gradient-boosted trees (XGBoost) with 200 estimators, max depth 3, learning rate 0.05, and column subsampling 0.8. The evaluation is repeated 5-fold cross-validation with 20 repeats (100 folds total), reporting cross-validated R^2, RMSE, and MAE. Random states are fixed (`random_state=42`) for reproducibility.

3.4 Causal identification

The causal pipeline targets the Average Treatment Effect on the Treated (ATT) of high Fearless Dominance on Professional Satisfaction.

Treatment. The treatment indicator is $T_i = \mathbb{1}\{\mathrm{FD}_i > \mathrm{P50(FD)}\}$ in the main analysis, with sensitivity at P67, P75, and P80. P50 (the median) balances the two arms most evenly and maximises analytic power.

Confounders. Eight observable confounders enter the propensity-score model: Age, Gender (male indicator), the five Big Five traits (Extraversion, Agreeableness, Conscientiousness, Emotional Stability, Openness), and Months in Job. Conditional Independence (CIA, Rosenbaum and Rubin, 1983; Imbens and Rubin, 2015) is assumed.

Propensity score. The propensity score is estimated by logistic regression (statsmodels `Logit`) on a design matrix of 16 parameters: an intercept, 8 linear terms (the seven continuous covariates plus the Gender indicator), and 7 quadratic terms for the continuous covariates only. We deliberately exclude pairwise interactions: an earlier 19-parameter specification with three theoretically motivated interactions (Extraversion × Emotional Stability; Age × Months in Job; Emotional Stability × Conscientiousness) increased McFadden pseudo-R^2 from 0.447 to 0.455 but did not improve the bimodality of the PS distribution; the parsimonious specification is preferred under $N \approx 200$ treated.

Caliper PSM. Following Austin (2011), we match 1:1 without replacement on the logit of the propensity score with caliper $0.2 \cdot \mathrm{SD}(\mathrm{logit}(\hat{p}))$. Treated units with no admissible match are dropped. Inference is via paired t-test combined with a percentile bootstrap of the matched-pair differences ($B = 500$); we deliberately do not re-fit and re-match the propensity score within the bootstrap, since this is invalid for matching estimators (Abadie and Imbens, 2008).

IPW. Inverse probability weighting uses the canonical ATT weights ($w_i = 1$ for treated, $w_i = \hat{p}_i/(1 - \hat{p}_i)$ for controls) after p_1/p_{99} trimming of the propensity score. Inference is by full-bootstrap with re-estimation of the propensity score in each replicate ($B = 500$).

Augmented IPW (AIPW). Following the Robins–Rotnitzky–Zhao formulation (Robins et al., 1995), the ATT is

$$\hat{\tau}_{\mathrm{AIPW}} = \frac{1}{n_1} \sum_i \left[T_i \left(Y_i - \hat{\mu}_0(X_i)\right) - (1 - T_i) \frac{\hat{p}_i}{1 - \hat{p}_i} \left(Y_i - \hat{\mu}_0(X_i)\right) \right], \tag{1}$$

where $\hat{\mu}_0(X_i)$ is an OLS prediction of the outcome on controls. We deliberately fit $\hat{\mu}_0$ as a *linear-only* function of the eight confounders, in contrast to the 16-parameter propensity-score model. Specification diversity between the two nuisance models is a precondition for the doubly-robust property: if both models share the same nonlinear misspecification, their errors covary and the asymptotic robustness disappears.

Bootstrap. Inference for IPW and AIPW uses $B = 500$ replicates with full re-estimation of both the propensity score and (for AIPW) the outcome model in each replicate. This captures the joint uncertainty of the two nuisance models. Caliper PSM uses paired-difference bootstrap on the matched sample, since re-matching across bootstrap replicates is invalid (Abadie and Imbens, 2008).

We report covariate balance via the Standardised Mean Difference (SMD) before and after adjustment, with $|\text{SMD}| < 0.10$ as the acceptable balance threshold. For caliper PSM, balance is unweighted (matched-pair means). For IPW, balance uses the same ATT weights. Entropy balancing (Hainmueller, 2012) is mentioned as an alternative we did not use; we rely on the diversity of three estimators rather than on a single best-balanced estimator.

3.5 Construct redundancy

The redundancy of FD with the Big Five is assessed in two complementary ways.

Linear redundancy. We regress FD on the Big Five and the three demographic controls (Age, Gender, Months in Job) and report R^2, adjusted R^2, the F-statistic, and standardised coefficients. We then estimate a parsimonious reduced model with only Extraversion and Emotional Stability and report the fraction of the full R^2 recovered. Variance Inflation Factors (VIF) for the full model assess multicollinearity among the predictors themselves (distinct from FD's redundancy with them).

Information-theoretic redundancy. We use the Kraskov–Stögbauer–Grassberger nearest-neighbour estimator (Kraskov et al., 2004) (KSG-1, $k = 3$, max norm) to compute three quantities. First, the univariate mutual information $I(\text{FD}; \text{BF}_k)$ for each Big Five trait separately, using the implementation in `sklearn.feature_selection.mutual_info_regression`. Second, the joint mutual information $I(\text{FD}; \text{BF}_{\text{full}})$ between FD and the five-dimensional Big Five vector, implemented in the script as a direct application of the KSG-1 estimator with multivariate Y. Third, the incremental MI of each Big Five trait given the other four, approximated as $\Delta I_k = I(\text{FD}; \text{BF}_{\text{full}}) - I(\text{FD}; \text{BF}_{\text{full}\setminus\{k\}})$, which identifies the unique contribution of each trait.

We complement these with a corrected redundancy ratio. The Shannon entropy of FD discretised into 10 quantile bins is $H(\text{FD}_{\text{disc}}) \approx \log_2(10) = 3.32$ bits, but mixing this discrete entropy with the continuous KSG mutual information produces an inconsistent ratio. The proper redundancy ratio uses the differential entropy of FD as the normaliser: under the standardisation $\text{FD} \sim \mathcal{N}(0, 1)$,

$$H_{\text{diff}}(\text{FD}) = \tfrac{1}{2} \log_2(2\pi e) \approx 2.05 \text{ bits}, \tag{2}$$

and the redundancy ratio is $1 - I(\text{FD}; \text{BF}_{\text{full}})/H_{\text{diff}}(\text{FD})$.

A final consistency check compares the rank-ordering of the Big Five by univariate MI against the rank-ordering by absolute standardised coefficient in the full OLS. Disagreement would indicate nonlinear dependencies that the linear model does not see.

4 Results

4.1 Replication of original findings

Table 1 reports Cronbach's α for the three constructs we replicate from the original. The values match the Eisenbarth et al. (2018) report to two decimal places. The Material Success composite

Table 1: Internal consistency, replication of Eisenbarth et al. (2018). Source: `output/replication_correlation_matrix.csv` and console output of `scripts/01_replication.py`.

Composite (# items)	α (this paper)	α (original)
Professional Satisfaction (3 items)	0.779	0.78
Material Success (3 items)	0.483	0.48
PPI-R Coldheartedness (5 items)	0.732	0.73

has $\alpha = 0.48$, below conventional reliability thresholds; this is a feature of the original instrument and is discussed in Section 6.

Table 2 reports zero-order Pearson correlations between PPI dimensions and the two outcome composites, alongside the Big Five Extraversion benchmark. Three patterns appear. First, the disaggregated PPI dimensions show opposite-sign effects on Professional Satisfaction: FD is positively associated ($r = +0.13$), SCI is negatively associated ($r = -0.14$), and CO is essentially null ($r = -0.05$). Second, the PPI Total score yields $r = -0.03$ on Professional Satisfaction (and +0.09 on Material Success), the direct consequence of cancellation between FD and SCI when summed. Third, Extraversion — a Big Five trait, not a psychopathic trait — shows the largest positive correlation with both outcomes ($r = +0.14$ and $r = +0.21$), foreshadowing the redundancy diagnostic of Section 4.5.

Table 2: Zero-order Pearson correlations between predictors and outcomes ($N = 439$). Stars denote $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***). Source: `output/replication_correlation_matrix.csv`.

Predictor	**Professional Satisfaction**	**Material Success**
FD	+0.13**	+0.15**
SCI	−0.14**	−0.01
CO	−0.05	−0.00
PPI Total	−0.03	+0.09*
Extraversion	+0.14**	+0.21***
Emotional Stab.	+0.09	+0.15**

The OLS replication of the original Table 4 is consistent with the published estimates: with Big Five and demographic controls included, the FD coefficient on Professional Satisfaction drops from +0.017 ($p = 0.009$) to +0.013 ($p = 0.215$), losing significance, while the SCI coefficient remains negative and significant (−0.023, $p = 0.031$). For Material Success, no PPI dimension is significant once Big Five controls are added; Extraversion is the dominant significant predictor ($\beta = +0.086$, $p < 0.001$).

4.2 Factor structure and aggregation bias

Before turning to the aggregation diagnostic, we briefly confirm the multidimensional structure of the PPI-R that justifies disaggregation. Bartlett's test of sphericity is highly significant ($\chi^2 = 4{,}821.99$, $p < 0.001$) and the Kaiser–Meyer–Olkin measure (KMO = 0.807) is in the "meritorious" range, confirming that the 40 PPI-R items are appropriate for factor analysis. The exploratory factor analysis with oblimin rotation (Figure 1) supports a multi-factor solution: the scree plot displays a clear elbow consistent with a 2-to-3 factor structure, and the loadings heatmap shows bipolar separation between items that load on the Fearless-Dominance factor and items that load on the Self-Centered-Impulsivity factor. The structural premise of the original instrument — that the 40 items measure orthogonal psychological dimensions rather than a single latent psychopathy construct — is empirically supported.

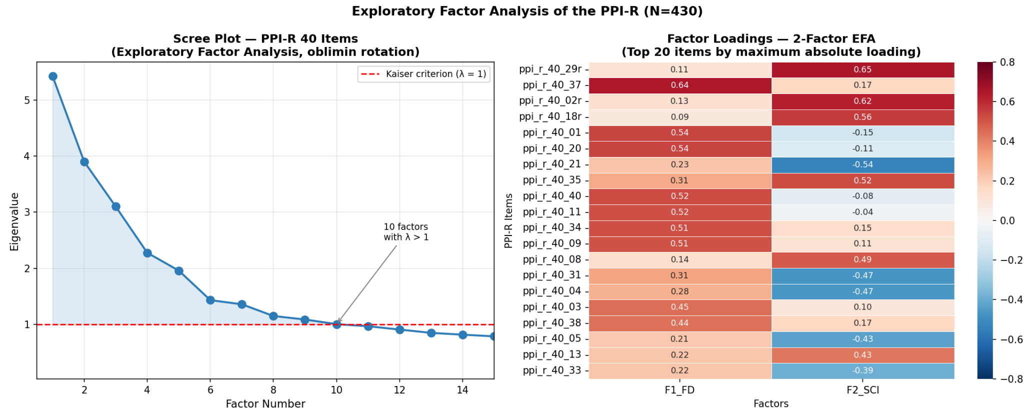

Figure 1: Exploratory Factor Analysis of the 40 PPI-R items (`factor_analyzer`, oblimin rotation, $N = 430$ complete cases). Left: scree plot with Kaiser criterion ($\lambda > 1$). Right: 2-factor loadings heatmap, top 20 items by maximum absolute loading. Source: `figures/fig11_efa_real.png`.

The aggregation bias itself is most clearly seen by visualising the component effects against the aggregate (Figure 2). The four-panel figure confirms that FD has a positive linear association with Professional Satisfaction, SCI a negative one, and the PPI Total score — being a sum of the two oppositely-signed components plus the null CO — yields a near-zero slope.

4.3 Predictive modelling

Table 3 reports the cross-validated R^2 and RMSE of OLS and XGBoost across four feature sets, on Professional Satisfaction. The point estimate for the best-performing OLS specification (the three PPI dimensions) is $R^2 = +0.005$; the best XGBoost specification has $R^2 = -0.069$. All R^2 values are within ± 0.04 of zero or negative; none of the models predicts Professional Satisfaction better than the unconditional mean by an economically meaningful margin.

Table 3: Cross-validated predictive performance for Professional Satisfaction (5-fold $\times$ 20 repeats; $N = 439$). Source: `output/formal_model_comparison.csv`.

Estimator	**Feature set**	CV R^2	CV RMSE	CV MAE
OLS	PPI Total only	−0.018	5.77	4.68
OLS	FD + SCI + CO	+0.005	5.70	4.66
OLS	Dimensions + Big Five	−0.011	5.75	4.69
OLS	Full model	−0.026	5.77	4.75
XGBoost	PPI Total only	−0.085	5.95	4.81
XGBoost	FD + SCI + CO	−0.069	5.91	4.80
XGBoost	Dimensions + Big Five	−0.073	5.91	4.82
XGBoost	Full model	−0.108	6.00	4.89

Two qualitative patterns are nevertheless robust. First, separating the three PPI dimensions yields a higher R^2 than aggregating them into the PPI Total — consistent with the aggregation-bias argument. Second, XGBoost performs strictly worse than OLS on every feature set, which is consistent with the absence of nonlinearities or interactions in the underlying predictive structure: the relationship, if any, is essentially linear and very weak.

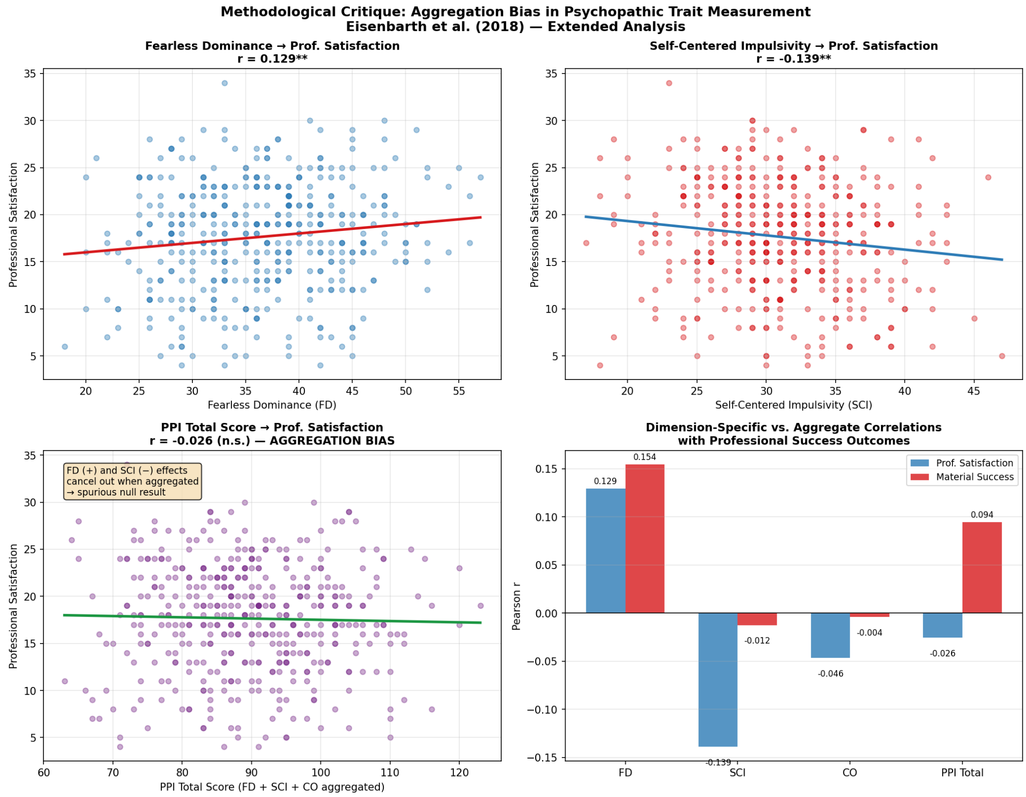

Figure 2: Aggregation bias diagnostic. Top-left: positive slope of Professional Satisfaction on FD ($r = +0.13$). Top-right: negative slope on SCI ($r = -0.14$). Bottom-left: near-zero slope on the PPI Total ($r = -0.03$), the cancellation of FD and SCI under aggregation. Bottom-right: bar comparison of dimension-specific vs. aggregated correlations across both outcomes. Source: `figures/fig7_aggregation_bias.png` produced by `scripts/02_extension.py`.

4.4 Causal estimates

Table 4 reports the main causal analysis at the median FD threshold (P50, $N_{\text{treated}} = 198$, $N_{\text{control}} = 211$, total $N = 409$ after listwise deletion). The three estimators yield positive but non-significant point estimates, ranging from +1.01 (caliper PSM) to +2.59 (IPW). The IPW p-value is marginally below the conventional 10% threshold ($p = 0.068$); neither caliper PSM ($p = 0.251$) nor AIPW ($p = 0.147$) clears the 10% threshold.

Table 4: Main causal estimates at FD > P50, outcome Professional Satisfaction. The caliper PSM column reports the matched-pair sample size; IPW and AIPW report the trimmed sample. Bootstrap with $B = 500$. Source: `output/att_main.csv`.

Estimator	**ATT**	95% **CI**	p**-value**	n used
Caliper PSM (1:1 NN, $0.2 \cdot \text{SD}(\text{logit PS})$)	+1.012	[−0.563, +2.674]	0.251	81 pairs
IPW (ATT weights, p_1/p_{99} trim)	+2.590	[−0.232, +5.178]	0.068	399
AIPW (Robins–Rotnitzky–Zhao, linear $\hat{\mu}_0$)	+2.092	[−0.966, +5.049]	0.147	399

The caliper PSM at P50 drops 117 of 198 treated units (59%) because no admissible control match exists within the caliper. This is a structural symptom of the redundancy diagnosed in Section 4.5: high-FD individuals tend to be high-Extraversion and high-Emotional-Stability simultaneously, and the joint distribution of Big Five traits among controls does not populate that region densely.

Figure 3 reports the Love plot of standardised mean differences before and after caliper matching at P50. Six of the eight covariates achieve the $|\text{SMD}| < 0.10$ threshold; two do not (Extraversion, SMD $= +0.18$; Emotional Stability, SMD $= +0.18$) — precisely the two Big Five traits most correlated with FD itself. The matching cannot balance covariates that are quasi-collinear with the treatment definition.

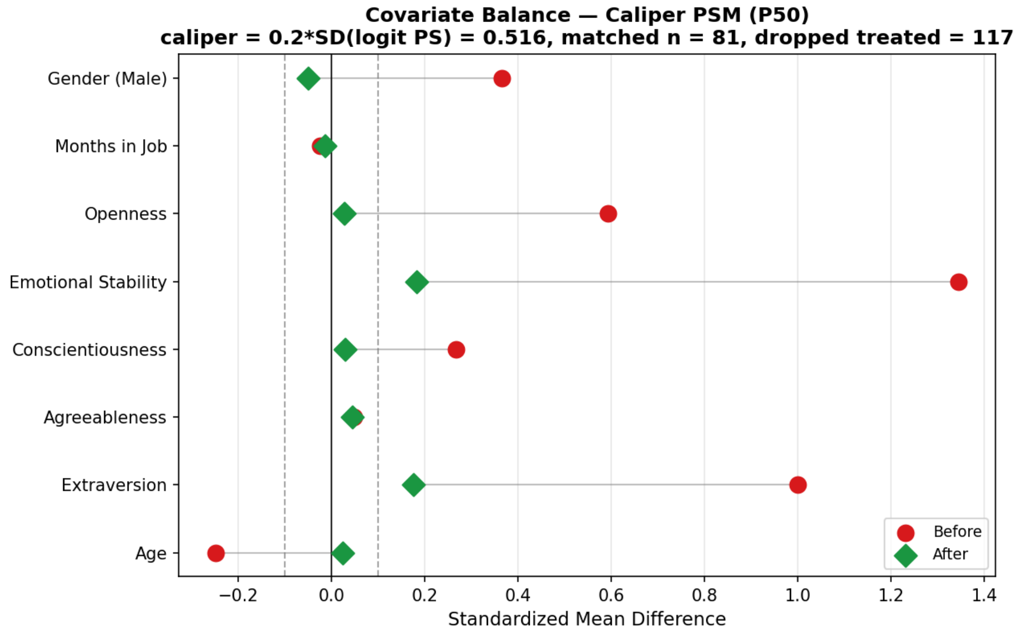

Figure 3: Covariate balance before and after caliper PSM at FD $>$ P50. Caliper $= 0.2 \cdot \text{SD}(\text{logit PS}) = 0.396$. Six of eight covariates achieve $|\text{SMD}| < 0.10$ (dashed lines). Source: `figures/love_plot_caliper.png`.

Figure 4 shows the equivalent diagnostic for IPW. Only Gender achieves the $|\text{SMD}| < 0.10$ threshold under ATT weights; the other seven covariates remain in the $[0.10, 0.34]$ band of weighted SMD. The pattern is informative: the ATT-weights re-balance the two strongest predictors of treatment (Extraversion, Emotional Stability) at the cost of inducing weighted imbalances in covariates with smaller marginal effects on the propensity score (Months in Job, Agreeableness, Age). The aggregate quality of IPW balance is therefore worse than caliper PSM by the $|\text{SMD}| < 0.10$ count (1/8 vs. 6/8) but with smaller maximum magnitudes overall.

Table 5 reports the sensitivity of the ATT to the choice of FD threshold and estimator (Figure 5 visualises this). No combination of threshold and estimator produces an ATT significant at the 5% level; the IPW $p = 0.068$ at P50 is the lowest. Across thresholds, the sign of the ATT switches between estimators and is not monotonic: the highest-FD subgroup (P80, $n_{\text{treated}} = 77$) yields an ATT close to zero (+0.09 AIPW, +0.14 IPW, +1.23 caliper PSM), the opposite of what one would expect under a genuine FD effect compounding with the strength of the treatment.

Table 5: Sensitivity of ATT to FD threshold across the three estimators. Bootstrap 95% CIs in brackets. Source: `output/att_sensitivity.csv`.

Threshold	Caliper PSM	IPW	AIPW
P50 ($n_t = 198$)	$+1.01\,[-0.56,\ +2.67]$	$+2.59\,[-0.23,\ +5.18]$	$+2.09\,[-0.97,\ +5.05]$
P67 ($n_t = 135$)	$-0.70\,[-2.42,\ +1.03]$	$-0.19\,[-2.28,\ +3.04]$	$-0.19\,[-3.10,\ +4.69]$
P75 ($n_t = 88$)	$+0.55\,[-1.40,\ +2.75]$	$-0.13\,[-1.99,\ +2.08]$	$-0.15\,[-2.34,\ +2.92]$
P80 ($n_t = 77$)	$+1.23\,[-0.59,\ +3.08]$	$+0.14\,[-1.54,\ +2.22]$	$+0.09\,[-1.58,\ +2.54]$

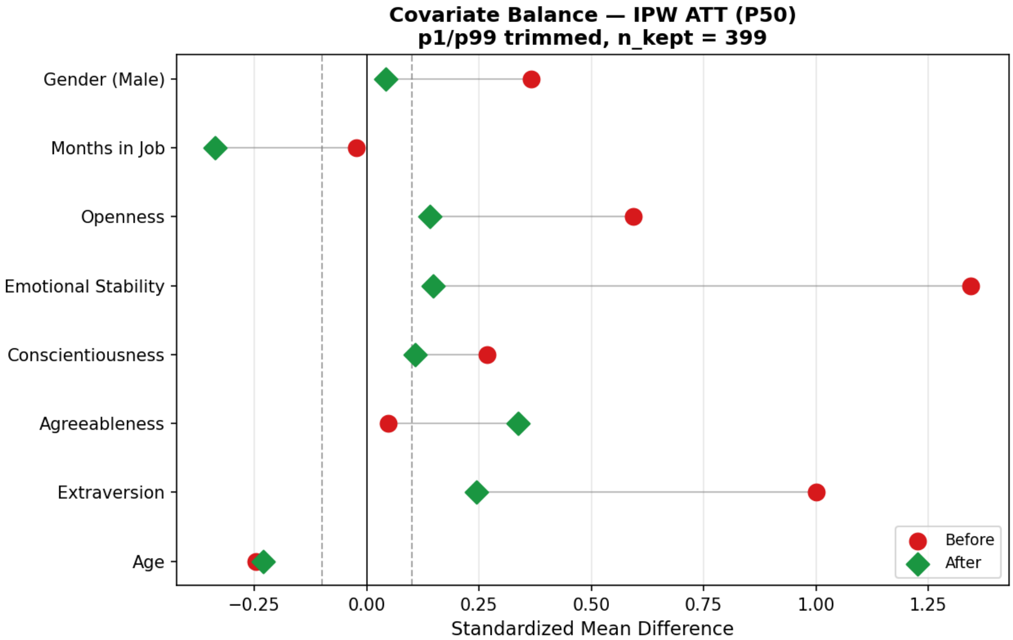

Figure 4: Covariate balance under IPW at FD > P50. ATT weights ($w = 1$ for treated, $w = \hat{p}/(1-\hat{p})$ for controls) after p_1/p_{99} trimming of the propensity score. One of eight covariates achieves $|\mathrm{SMD}| < 0.10$. Source: `figures/love_plot_ipw.png`.

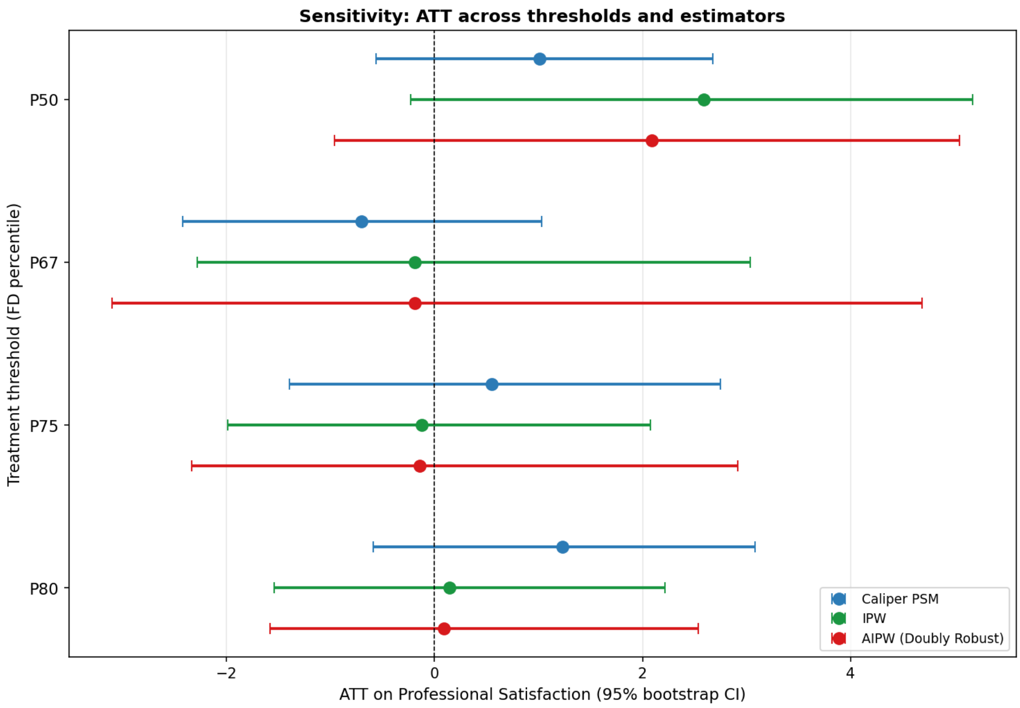

Figure 5: Sensitivity of the ATT across four thresholds and three estimators, with 95% bootstrap CIs ($B = 500$, full propensity-score re-fit per replicate for IPW and AIPW; matched-pair bootstrap for caliper PSM). Source: `figures/sensitivity_forest_plot.png`.

4.5 Construct redundancy

Table 6 reports the linear and information-theoretic redundancy diagnostics. Panel A shows that the regression of FD on the Big Five plus Age, Gender, and Months in Job yields $R^2 = 0.616$

($N = 409$, $F = 80.11$, $p < 10^{-77}$). The reduced model with only Extraversion and Emotional Stability captures $R^2 = 0.539$, or 87.5% of the full-model explanatory power. The standardised coefficients confirm that Emotional Stability ($\beta = +0.51$, $p < 10^{-36}$) and Extraversion ($\beta = +0.30$, $p < 10^{-16}$) are the dominant predictors of FD; Agreeableness and Conscientiousness are not significant. VIF among the predictors themselves are all below 1.5, indicating that the redundancy is between FD and the joint Big Five vector, not multicollinearity within the Big Five.

Panel B reports the information-theoretic counterpart. Univariate mutual information confirms the linear pattern: Emotional Stability contributes 0.47 bits, Extraversion 0.17 bits; the remaining three Big Five traits contribute essentially zero. The joint MI of FD with the five-dimensional Big Five vector is 0.93 bits. The incremental MI of each Big Five given the other four is positive only for Emotional Stability (+0.27 bits) — the other four show zero or negative incremental contribution due to KSG estimator variance at $N = 409$, but the qualitative point is unambiguous: only Emotional Stability adds unique information once the others are known. The differential entropy of standardised FD is $H_{\text{diff}}(\text{FD}) = \frac{1}{2}\log_2(2\pi e) = 2.05$ bits; the corrected redundancy ratio is $1 - 0.93/2.05 = 0.55$. The Big Five jointly accounts for 45% of FD's information content; the remaining 55% is idiosyncratic variance or measurement noise, with no detectable structure given the Big Five.

The rank-ordering of the Big Five by univariate MI (Emotional Stability, Extraversion, Openness, Conscientiousness, Agreeableness) matches the rank-ordering by absolute standardised coefficient *position by position* (5/5). This is direct evidence that the FD–Big-Five relationship is well-described by a linear model: the information-theoretic estimator does not reveal any nonlinear structure that the OLS would miss.

Table 6: Construct redundancy of FD with Big Five. Panel A: linear diagnostics ($N = 409$). Panel B: information-theoretic diagnostics (KSG-1, $k = 3$). Sources: `output/fd_redundancy_analysis.csv` and `output/fd_redundancy_information.csv`.

Panel A: Linear redundancy	
OLS full R^2 (FD $\sim$ Big Five + Age + Gender + Months in Job)	0.616
OLS reduced R^2 (FD $\sim$ Extraversion + Emotional Stability)	0.539
Reduced fraction of full R^2	87.5%
F-statistic (full model)	80.11 ($p < 10^{-77}$)
Pearson r (FD, Emotional Stability)	+0.66***
Pearson r (FD, Extraversion)	+0.49***
Standardised β on Emotional Stability	+0.51 ($p < 10^{-36}$)
Standardised β on Extraversion	+0.30 ($p < 10^{-16}$)
Maximum VIF among Big Five (no multicollinearity within BF)	1.39
Panel B: Information-theoretic redundancy (bits)	
$H_{\text{diff}}(\text{FD})$ (Gaussian, $\sigma = 1$): $\frac{1}{2}\log_2(2\pi e)$	2.05
Joint MI $I(\text{FD}; \text{BF}_{\text{full}})$ (KSG-1)	0.93
Univariate MI: Emotional Stability	0.47
Univariate MI: Extraversion	0.17
Univariate MI: Openness	0.15
Univariate MI: Conscientiousness	0.06
Univariate MI: Agreeableness	0.00
Incremental MI of Emotional Stability (given other 4 BFs)	0.27
Redundancy ratio $1 - I(\text{FD}; \text{BF})/H_{\text{diff}}(\text{FD})$	0.55
MI ranking vs. $\lvert\beta\rvert$ ranking agreement	5/5

5 Discussion

5.1 Why the three problems are connected

The three problems we diagnose — aggregation bias, construct redundancy, and null causal effect — are not independent. They share a common architecture.

Aggregation bias is the symptom that motivates the rest. The PPI Total yields a near-zero correlation with Professional Satisfaction ($r = -0.03$), which would lead a naive analyst either to dismiss the psychopathy–success question or to seek nonlinearities. We instead disaggregate, recovering FD's positive correlation ($r = +0.13$) and SCI's negative correlation ($r = -0.14$). At this point a typical empirical paper would report the FD effect on Professional Satisfaction and stop.

The redundancy diagnostic intervenes here. Once we observe that $R^2(\text{FD} \sim \text{Big Five}) = 0.62$, with Extraversion and Emotional Stability accounting for 87.5% of that — and the information-theoretic check confirming the linear pattern — the question of FD's incremental causal contribution becomes substantively unanswerable from observational data: the construct of interest is near-linear in the controls. This is precisely what we observe in the causal pipeline. The caliper PSM at P50 drops 59% of treated units because the treatment region has no comparable control region in the joint Big Five space; the IPW weights re-balance the two strongest predictors at the cost of imbalance on others; the AIPW estimator splits the difference between the two and yields a wider confidence interval. None of the estimators identifies a significant ATT.

The honest reading is therefore not "FD has no causal effect" in the strong sense — the data do not allow that strong inference. The honest reading is that the research design is uninformative for the FD effect because the treatment is too close to a deterministic function of the controls.

5.2 Implications for the psychopathy literature

A non-trivial fraction of empirical work in personality psychology and labor economics reports associations between PPI Total or FD and career outcomes without controlling for Big Five Extraversion and Emotional Stability. Our diagnostic suggests that any such association is likely to disappear or diminish substantially once the Big Five controls are included. This is not a novel observation in personality theory — Miller and Lynam (2012) have argued the redundancy case for the PPI's lower-order factors — but the joint diagnostic combining linear regression and KSG mutual information is, to our knowledge, new for this dataset. The practical recommendation is that PPI Total or FD should not be reported as a predictor of professional outcomes without an explicit Big Five control specification, and that the reduction of the FD coefficient when Big Five controls are added is a structural feature of the construct, not a confound to be explained away.

The Triarchic conceptualisation of psychopathy (Patrick et al., 2009), which decomposes psychopathy into Boldness, Meanness, and Disinhibition, partly anticipates this concern: the "Boldness" factor is widely understood to overlap heavily with adaptive personality traits (Extraversion, low Neuroticism), and our findings reinforce that overlap empirically.

5.3 Implications for labor economics

The labor-economics literature on non-cognitive skills, surveyed by Borghans et al. (2008) and consolidated by Heckman and Kautz (2012), establishes that personality traits predict earnings, employment, and occupational attainment at magnitudes quantitatively comparable to cognitive ability for many outcomes, and that ignoring them generates omitted-variable bias in returns-to-schooling estimates. Within this literature the Big Five is the canonical operationalisation. Mueller and Plug (2006), using longitudinal earnings data, estimate non-trivial returns to one standard deviation of each Big Five trait, with substantial gender heterogeneity in which traits carry the strongest returns. Against this baseline, any claim that a distinct "psychopathy premium" exists in professional outcomes is implicitly a claim that some FD-related trait variation remains *after* conditioning on the Big Five and contributes independently to compensation or satisfaction. The empirical bar is therefore not whether FD correlates with success in raw data — Eisenbarth et al. (2018) confirm it does — but whether FD adds information that the Big Five does not already carry.

The diagnostic of Section 4.5 answers this question negatively in the present data. FD is not statistically separable from Extraversion combined with Emotional Stability: the linear R^2 of FD on the Big Five and demographics is 0.62, the corrected redundancy ratio $1 - I/H_{\text{diff}}$ is 0.55, and the rank-orderings produced by the linear model and by the KSG mutual-information estimator agree on all five Big Five positions. A premium attributable to FD, in this sample, collapses mechanically into the premium already documented by the Big Five literature for Extraversion and Emotional Stability. There is no residual variation in FD — after conditioning on the Big Five — that could be reassigned to a distinct "successful psychopathy" construct.

The policy implication of this redundancy depends on which narrative one accepts. The first narrative, "psychopathy is rewarded in the workplace", treats FD as a distinct trait cluster of dark personality features that labour markets have mis-priced, and suggests a policy response of re-designing hiring and promotion processes to penalise it. The second narrative, "social confidence and emotional resilience are rewarded", treats the same statistical pattern as the conventional return to non-pathological personality already documented in the Big Five literature (Borghans et al., 2008; Mueller and Plug, 2006), and suggests the more conventional policy implication of recognising personality as a labor-market input alongside cognitive ability and human capital. Our null causal results, conditional on the redundancy, suggest the second narrative is the more parsimonious description of the data.

6 Limitations and threats to identification

This is a working draft (v0.1). Several limitations apply.

Material Success reliability. The Material Success composite has Cronbach's $\alpha = 0.48$, well below conventional psychometric thresholds. We retain Material Success only for the descriptive replication (Table 2); the predictive, causal, and redundancy analyses use Professional Satisfaction ($\alpha = 0.78$) as the primary outcome. This restricts the scope of the conclusions to Professional Satisfaction and leaves Material Success unaddressed by our extension.

Predictive performance is near zero. The cross-validated R^2 of every model in Table 3 is within a few percent of zero or negative. This is a substantive limit: even if the FD effect were real, no model in our specification space predicts Professional Satisfaction well from any combination of psychopathic traits, Big Five traits, and demographics. The variance explained by all available predictors is small relative to the residual.

Conditional Independence Assumption. Identification of the ATT requires that, conditional on the eight observed covariates, treatment assignment is independent of potential outcomes. We do not observe cognitive ability, socioeconomic background, sectoral heterogeneity, or industry-specific compensation structures. Any of these may correlate with both self-rated FD and professional success, violating CIA in directions we cannot bound.

Reverse causality. Professional success may causally elevate self-reported Fearless Dominance over time. The PPI-R is self-administered at a single time point; we cannot separate "confident successful person" from "successful person who became more confident as a consequence of success."

Self-report measurement error. Both the predictors (PPI-R, Big Five) and one of the outcomes (Professional Satisfaction) are self-reported. Self-rating bias may inflate the apparent correlation between personality and self-perceived success. The Material Success composite is partially anchored on objective indicators (annual salary band, promotion frequency) but its reliability is low.

Non-experimental treatment. The treatment "high FD" is defined by thresholding a continuous personality measure. This is not a treatment in the experimental sense; the comparison group is a post-hoc partition of the same population on a self-reported trait that, as Section 4.5 establishes, is near-linear in the Big Five controls.

Common-support failure. The caliper PSM drops 59% of treated units at P50. The reported caliper-PSM ATT is therefore restricted to the 81 treated units who do find an admissible match; the 117 units without a match represent a region of the covariate space where causal inference under CIA is not feasible from this sample.

7 Conclusions

Three convergent lines of evidence support the view that Fearless Dominance is not empirically separable from Big Five Extraversion and Emotional Stability in this sample. First, the original PPI Total correlation with Professional Satisfaction ($r = -0.03$) is a spurious null produced by the cancellation of opposite-sign FD (+0.13) and SCI (−0.14) component effects. Second, FD is captured at $R^2 = 0.62$ by the Big Five plus demographics, with 87.5% of that recovered by Extraversion and Emotional Stability alone, and the information-theoretic redundancy ratio is 0.55 — of FD's information content, 45% is shared with the Big Five and the remainder is idiosyncratic. Third, three causal estimators applied at four FD thresholds yield no ATT significant at 5%, and the caliper PSM at the median threshold drops 59% of treated units for lack of admissible matches.

The contribution is methodological. The combination of (i) an aggregation-bias diagnostic, (ii) a corrected redundancy ratio that combines the KSG mutual information with Gaussian differential entropy, and (iii) a causal pipeline with three estimators and four thresholds gives a transparent template for assessing whether a candidate construct adds incremental information to a baseline of established personality dimensions. Applied here, the template suggests that previously reported FD effects on professional outcomes are plausibly attributable to the Big Five components on which FD loads.

Future work suggested by this analysis includes longitudinal designs to address reverse causality directly — the available cross-section cannot — and instrumental-variable strategies for FD that exclude the Big Five from the first stage. Both are demanding empirical projects given the apparent linearity of FD in the Big Five; the exclusion restriction in particular requires a source of variation in FD that is plausibly orthogonal to Extraversion and Emotional Stability, which is not obviously available in standard personality datasets. A complementary direction is the application of the same three-pronged diagnostic to other "successful psychopathy" operationalisations in labour-economics datasets, to assess whether the redundancy with Big Five generalises beyond the PPI-R.

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